

6	(a) (i)	coherent: constant phase difference	B1	
		interference is the (overlapping of waves and the) sum of/addition of displacement of two waves	B1	[2]
	(ii)	wavelength = 3.2 m (allow ± 0.05 m)	M1	
		$f (= v / \lambda = 240 / 3.2) = 75 \text{ Hz}$	A1	[2]
	(iii)	90° (allow $\pm 2^\circ$) or $\pi/2$ rad	A1	[1]
	(iv)	sketch has amplitude 3.0 ± 0.1 cm	M1	
		correct displacement values at previous peaks to produce correct shape	A1	[2]
	(b) (i)	$\lambda = ax/D$	C1	
		$x = (546 \times 10^{-9} \times 0.85) / 0.13 \times 10^{-3} (= 3.57 \times 10^{-3} \text{ m})$	C1	
		$AB = 8.9 (8.93) \times 10^{-3} \text{ m}$	A1	[3]
	(ii)	shorter wavelength for blue light so separation is less	B1	[1]